

- 5 (a) (i) To channel / strengthen / increase / concentrate the magnetic field produced by the 2 wires through the sensing coil.
- (ii) When there is a fault, the current in the 2 wires between the sensing coil will be unequal. The net magnetic flux through the sensing coil is no longer zero. (Since the current is alternating,) the changing magnetic flux will induce an emf, according to Faraday's law, and trigger the circuit breaker.
- (iii) Increase the number of turns of the sensor coil.
- (b) (i)
$$\begin{aligned} \text{Ave } \epsilon &= \frac{\Delta N B A}{\Delta t} \\ &= N B \frac{\Delta A}{\Delta t} \\ &= 1 \times (25.0 \times 10^{-3}) \times \frac{\pi \times 0.0100^2}{50.0 \times 10^{-3}} \\ &= 1.57 \times 10^{-4} \text{ V} \end{aligned}$$
- (ii) According to Lenz's law, as the loop shrinks, the direction of the induced EMF is such that it will oppose the motion by causing an outward radial force on the loop. Hence, by Fleming's left hand rule, induced current will flow from A to B and hence, B is at a higher potential than A.
- (c) (i) frequency = 100 Hz, period = 10 ms
Hence, as 2 cm represents 10 ms, the time base setting is 5 ms cm^{-1}
- (ii)
$$\begin{aligned} V_{\text{rms}} = \sqrt{\langle V^2 \rangle} &= \sqrt{\frac{0.75^2 \times 1 + (-0.5)^2 \times 3}{4}} \\ &= 0.573 \text{ V} \end{aligned}$$

